Advanced statistics

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Feature scaling

Age	YOE	salary
25	2	1000
35	10	5000
40	15	7000
38	13	5400
53	22	1200
42	12	4500

different scale

same scale

- In statistical tests, machine learning, columns having higher scales can dominate the analysis which wrongly affects the results

- hence we try to match scales of all columns/features using feature scaling.

- There are two popular feature scaling techniques

1) min-max scaling

feature/column = [5, 10, 15, 25, 35]

min = 5 max = 35

$$x' = \frac{x - min}{max - min}$$

$$X = [5, 10, 15, 25, 35]$$

$$\text{max} - \text{min} = 35 - 5 = 30$$

$$X' = \left[\frac{5-5}{30}, \frac{10-5}{30}, \frac{15-5}{30}, \frac{25-5}{30}, \frac{35-5}{30}\right]$$

$$X' = \left[0, \frac{5}{30}, \frac{10}{30}, \frac{20}{30}, 1\right]$$

min-max scaling makes the transformed data lie in the range / scale of 0-1

2) Z-score scaling / standardization

$$x' = \frac{x - \mu}{\sigma}$$

$$X = [5, 10, 15, 25, 35]$$

$$\mu = 18$$

$$\sigma$$

$$X' = \left[\frac{5-\mu}{\sigma}, \frac{10-\mu}{\sigma}, \frac{15-\mu}{\sigma}, \frac{25-\mu}{\sigma}, \frac{35-\mu}{\sigma}\right]$$

mean of $X' = 0$

- After standardization, data becomes zero centered ie mean = 0. This zero centered data works well with ML and DL models.

- Scale of the features also becomes similar.

When feature scaling is done, same technique is applied on all features i.e. either min-max scaling or standardization.

Inferential Statistics

Q. Whats the average income of an Indian citizen?

$\longrightarrow$

Method 1

1.5 B population $\longrightarrow$ get everyone's income and compute avg

Method 1 has lot of resource constraints - time, money, etc.

1.5 B population $\rightarrow$ mean income $= \mu$

$\downarrow$

sample of 10k people (good sample) $\longrightarrow$ mean income $= \mu'$

* According to inferential stats, if we take a good enough sample of the population then $\mu' \approx \mu$

	Population	Sample
size	$N = 1.5B$	$n = 10k$
mean	μ - population mean	$\bar{x}$ - sample mean
std dev	σ - population std dev	s - sample std dev

In inferential stats, we try to <u>estimate</u> <u>population parameters</u> (μ, σ) using <u>sample statistics</u> $(\bar{x}, s)$

Descriptive stats → applied on population → 100% accurate

Inferential stats → applied on sample → not 100% accurate estimate

process of estimating population parameters using sample statistics

1) collect the sample
2) calculate sample statistics ($\bar{x}$, s)
   this step is called point estimation
3) confidence interval estimation

Avg income of indian person

μ → population mean N = 1.5B

1) sample collection from population
   n = 10k

2) compute $\bar{x}$, s
   $\bar{x}$ = 12,000
   s = 5000

3) confidence interval estimation

- we are 95% sure that $\boxed{\bar{x} \approx \mu}$

range for $\mu = [\bar{x} - k\sigma, \bar{x} + k\sigma]$

Sampling Techniques

- sample has to be a good representation of population.
- sample should have approximately all properties of population.
- sample should not have any <u>bias</u>.

eg. avg weight of 10th class students

sample 1000 students → 800 M + 200 F

↓

sample is biased towards male students

1) Convinient Sampling

- sample is taken with convinience of resources.
- this technique has high chance of introducing bias

eg.

taking survey at railway station rather than going to market.

2) Quota Sampling

- proportions of certain property of population is maintained.

eg. maintain proportion of genders

60% M + 40% F

maintain proportion of age range

1 - 20	20 - 40	40-60	60+
15%	40%	25%	20%

maintain proportion of source

10 lac iron rods produced by 4 machines

M1 - 2.5 lac
M2 - 2.5 lac
M3 - 2.5 lac
M4 - 2.5 lac

} sample of 1k iron rods → 250 M1, 250 M2, 250 M3, 250 M4

- issue with this technique is, we might not know quota of property in some cases.

3) <u>Systematic Sampling</u>

- collecting sample based on a fixed strategy
- it might not be suitable for some use cases

eg. we want to take sample of 100 rods manufactured by a machine. in a day machine produces 10000 rods. we take every 100th rod in our sample.